Physical Activity-Based Prediction of Hyperglycemia and Hypoglycemia in Youth with Type 1 Diabetes

Mizzou

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Introduction

Type 1 diabetes (T1D):

Type 1 Diabetes (T1D) is a chronic autoimmune condition in which the body's immune system attacks the insulin-producing beta cells in the pancreas, leaving individuals unable to produce insulin. As a result, people with T1D require daily insulin therapy through injections or insulin pumps to regulate their blood glucose levels. Approximately **1.4 million people in the U.S.** are living with T1D—comparable to the population of San Antonio, Texas.

Unique Challenges for Youth with T1D:

Physiological and emotional development

- Peer pressure and social dynamics may influence lifestyle choices, increasing stress and impacting blood sugar control.
- Physical activities cause unpredictable blood sugar swings.

Health Consequences

- Serious complications like Diabetic Ketoacidosis (DKA)
- Hypoglycemia and hyperglycemia
- Long-term issues: cardiovascular, nerve, and eye diseases

Dataset

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Questionnaires

- Participants questionnaires
  Eg: Barriers to physical activity, Hypoglycemia awareness
- Parent questionnaires

BANT App

- Manually entered activity, insulin dosage, and meal data.
- MDI users record their insulin dosage on the app.

Diabetes Device

- Continuous Glucose Monitoring data
- Insulin Dosage
  Eg: Tandem t:slim X2, Insulet Omnipod dash

Garmin Vivosmart 4

Fitness Wearables

- Heart rate
- Sleep
- Stress
- Physical activity
- Motion
- Intensity

Methodology

Data Integration:

Aligned timestamps across CGM data, physical activity logs (Garmin and Bant App), and contextual information to create a unified dataset.

Event Identification:

Selected exercise events lasting ≥ **5 minutes** for inclusion.

Labeling Outcomes:

- Defined a **2-hour post-exercise window** to observe glycemic outcomes.
- Labeled events as **hypoglycemia or hyperglycemia** if the condition lasted ≥ **15 minutes** within that window.

Input Window Construction:

Extracted **2-, 4-, and 6-hour windows** of CGM data **prior to each exercise** event for model input.

Modeling Approaches:

- MERF (Mixed Effects Random Forest), LSTM

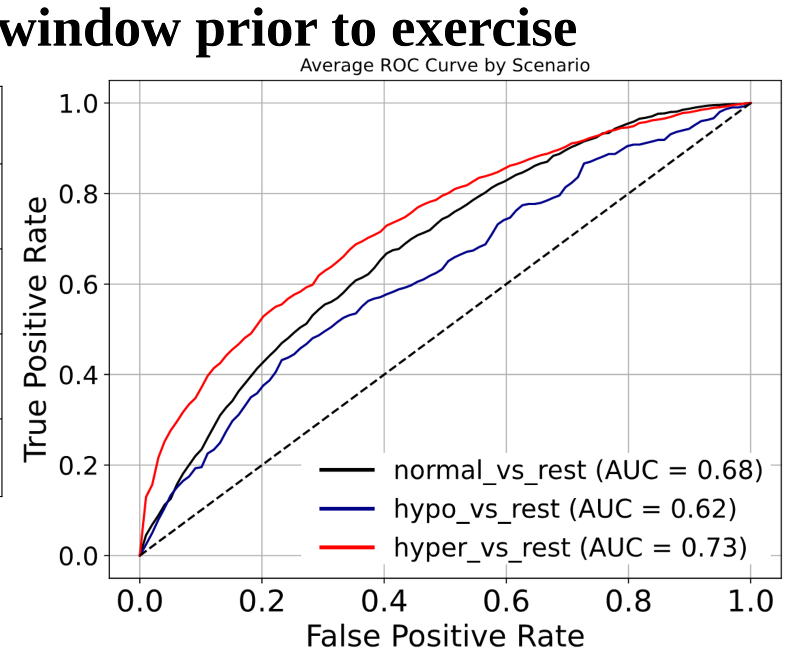
Results

MERF- One vs. Rest strategy for class-specific prediction.

- After downsampling, using a one-vs-rest approach, the MERF model performed best when using a **6-hour window prior to exercise**

Models \ Metrics	Precision	Recall	Accuracy
Hypo vs. Rest	0.12	0.54	0.64
Normal vs. Rest	0.60	0.61	0.63
Hyper vs. Rest	0.65	0.60	0.67

Mixed-Effects Random Forest 6-hour



LSTM- multi-class classifier

- **6-hour window gives the best result** among the combinations.
- The model performs best at detecting normal cases but struggles especially with hypoglycemia.

Resampling Methods	2hr	4hr	6hr
Downsampling	0.39	0.39	0.42
Oversampling	0.44	0.43	0.49

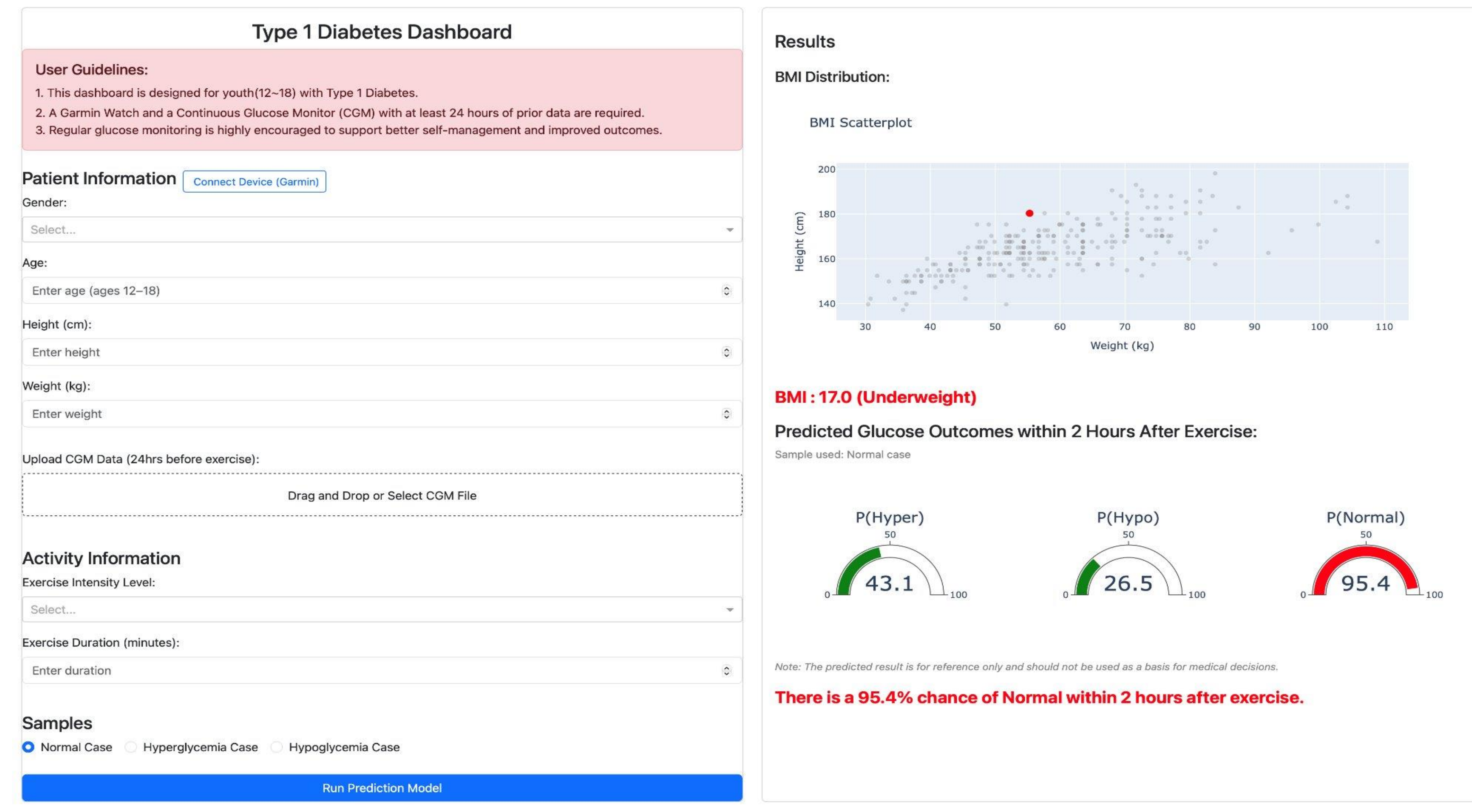
Condition	Precision	Recall	Accuracy
Hypoglycemia	0.55	0.20	0.29
Normal	0.51	0.64	0.57
Hyperglycemia	0.44	0.55	0.48

Overall Accuracy: 0.49

Application and Conclusion

Dashboard

- An interactive dashboard designed specifically for youth aged 12 to 18 with Type 1 Diabetes.
- The dashboard leverages a MERF model using a one-vs-rest strategy to predict post-exercise glycemic outcomes—hypoglycemia, normoglycemia, or hyperglycemia—based on individual glucose history and activity data.



Conclusion

- Model selection and input window design should align with the specific prediction objective—whether prioritizing overall classification or targeted detection (e.g., hypoglycemia events).
- Extending the input time window prior to physical activity improved performance in both models by capturing longer-term glucose trends, with longer windows consistently enhancing prediction accuracy.

Future Work

- Extending the input time window further before activities may help the LSTM model capture longer-term glucose trends and improve prediction accuracy.
- Incorporate behavioral studies to better understand individual patterns and enhance personalized glucose management.